



EXPERIMENT NO. 01

STUDY OF STRUCTURE- PROPERTY RELATIONSHIP OF MATERIALS FROM DIFFERENT VIEWPOINTS

Submitted by,

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Abstract:

The purpose of this experiment was to examine various materials and their structure and properties from different perspectives. Each team received a sample of a different material and performed tests to determine its properties. Our team, group 02, was assigned an Al specimen. We first measured the gauge length and diameter and carried out a tensile test. Then we cut a 10mm piece from the end of the specimen and conducted a microstudy. Finally, we did a hardness test and computed the relevant parameters using the appropriate formulas. We calculated the differences between the values came out from graph and hardness test.

Objectives:

Comparison of ferrous (TMT, non-TMT steels, cast irons, etc.) and non-ferrous alloys (brass, aluminum, etc.) obtained from tensile test and calculated knowing microstructures and/or converted from hardness data.

Introduction:

We had given samples of-

TMT Steels: Thermo mechanically treated steel known as TMT steel can be described as a new-generation-high-strength steel having superior properties such as weldability, strength, ductility, and bendability meeting the highest quality standards at the international level. (gopinath, n.d.)

Non-TMT Steels: These steels are not thermos mechanically treated. It generally has high carbon content which increases its strength but reduces its ductility.

Cast Irons: Cast iron, an alloy of iron that contains 2 to 4 percent carbon, along with varying amounts of silicon and manganese and traces of impurities such as sulfur and phosphorus. (Encyclopedia Britannica, n.d.)

Aluminium alloy: An aluminium alloy is an alloy in which aluminium (Al) is the predominant metal. The typical alloying elements are copper, magnesium, manganese, silicon, tin, nickel and zinc.

Brass: Brass is a mixture of copper (Cu) and zinc (Zn), which can have different ratios to change its colour and its physical, electrical, sound, and chemical characteristics, but usually copper is the main component.

As we are group-02, Al alloy was given to us. We conducted tensile test, microstudy, hardness test to our sample. Calculated the parameters using those experimental value-

$$1. \quad YS = 0.383 \cdot HB - 182.3$$

$$2. \quad UTS = 0.247 \cdot HB + 113.1$$

Where HB is Brinell hardness. (David Merayo, 2020)

Further we have calculated the strength of our sample using the following formulas:-

$$UTS = \sigma_{gb} + \sigma_p + \sigma_{ss} + \tau_D$$

$$1. \quad \sigma_{gb} = \sigma_{\alpha} + k_y d^{-1/2}, \text{ where } d \text{ is the average grain size.}$$

$$2. \quad \sigma_p = 362.7 f_1^{1/2} r_1^{1/2} + \frac{0.1}{\pi} \frac{M G b}{\sqrt{1-\nu}} \frac{\ln\left(\frac{4r_2}{b}\right)}{r_2 \left(\sqrt{\frac{\pi}{4f_2}} - 1\right)}$$

$$3. \sigma_{ss} = \sum AC_0^{2/3}$$

$$4. \tau_D = \frac{BGb}{\lambda}$$

$$5. \lambda = 4r_2 \left(\sqrt{\frac{\pi}{4f_2}} - 1 \right) \text{ (Ruishuai Gao a b c, May–June 2022)}$$

A microstructural analysis of an aluminium sample alloyed with magnesium was performed, which presented a high level of difficulty. The results and implications of this analysis were further examined in detail.

Tools and equipment:

- 1)Aluminum alloy
- 2)Universal testing machine
- 3)Vernier caliper
- 4)Hacksaw
- 5)Grinding machine
- 6)Emery paper
- 7)Wet polishing machine
- 8)Microscope
- 9)Brinell Hardness Tester

Procedure:

To conduct the experiment we followed the following procedures:-

- 1)First, we obtained the sample (Al) from the instructor. We then took it to the material testing lab, where we conducted a tensile test on our specimen using the UTS machine.
- 2)After the test, we cut a 10mm section from the end of our specimen, which was fixed to a vice using a hacksaw.
- 3)We then transported the cut section to the microstructure study lab and performed 150 emery paper grinding with a machine.
- 4)Subsequently, we manually conducted emery paper grinding with grit sizes 320, 600, 800, and 1200.
- 5)Further, we polished the specimen using a wet polishing machine.



6) Upon completing the steps, we immediately and gently wiped the surface with acetone to prevent the formation of an oxide layer.

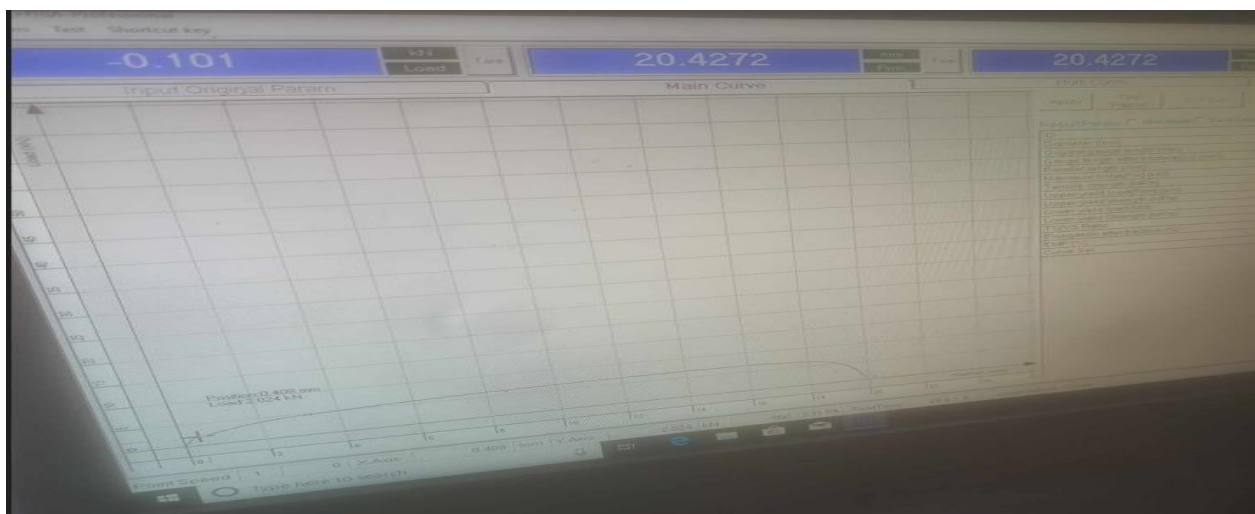
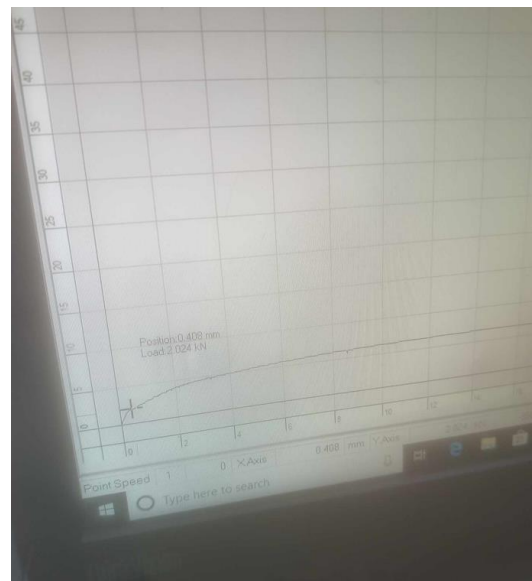
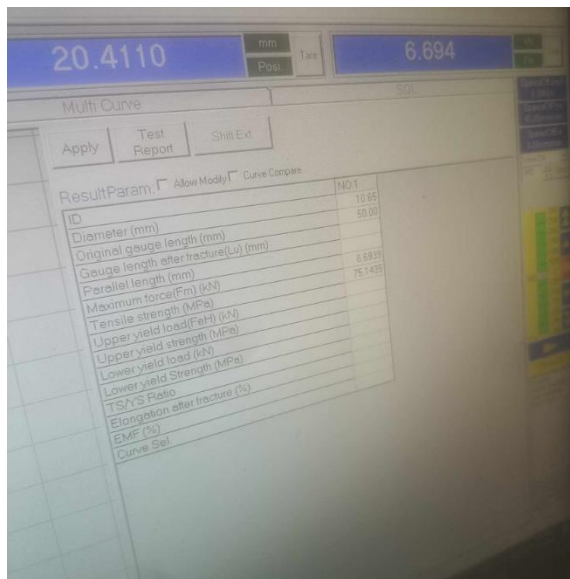
7) The next day, we carried out etching using a reagent consisting of 5% HF, 20 mL distilled water, 20 mL HNO_3 , and 20 mL HCl.

8) Finally, we placed the specimen under the microscope and captured its microstructure for further calculations.

9) After getting a microscopic structure, we took the sample to the 'Materials Characterization Lab' to calculate the hardness of the specimen. We did the Brinell Hardness Test.

10) We placed the specimen in the Brinell Hardness Test machine. A 500 kg load was applied for 10 seconds in the specimen. As a result, a circular indentation was created in the specimen.

11) After that we took the sample back to the Metallography Lab to measure the diameter of the indentation.



Data and Microstructure:

Diameter=10.65mm

Original gauge length =50mm

Gauge length after rupture=65mm

Tensile strength=75.1435MPa

Maximum force=6.6939kN

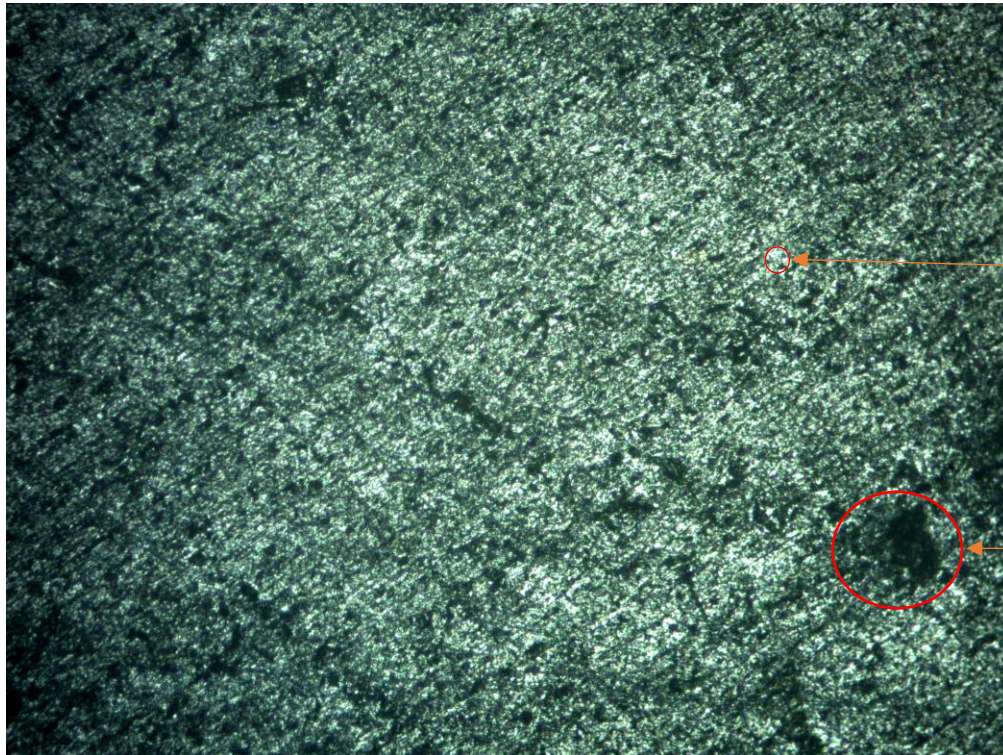
Tensile strength=75.1435MPa

Brinell hardness test data,

Small diameter=6.37mm

Big diameter=10mm

Load=500kgf



dendrites

Porosity

Magnification:100X

Etching reagent: 5%HF+20mL distilled water+20mL

HNO_3 +20mLHCl



Corroded surface because of multiple times operations

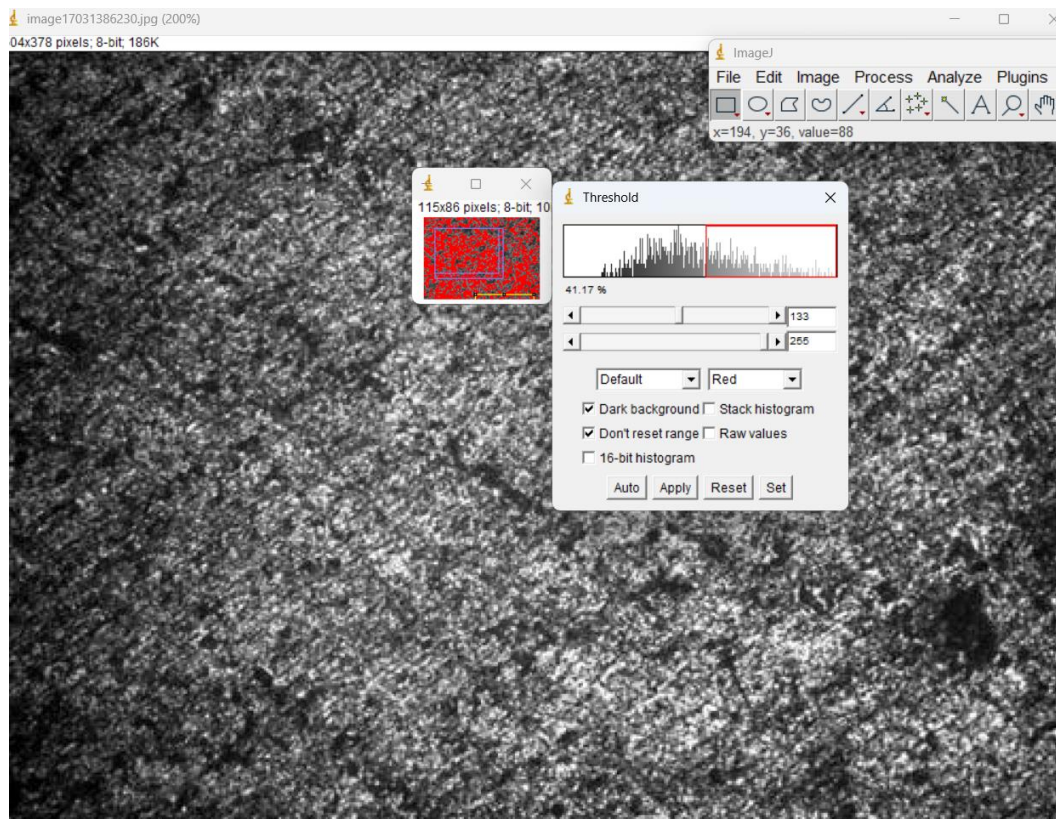
Dendrites

Porosity

Magnification:500X

Etching reagent: 5%HF+20mL distilled water+20mL HNO_3 +20mLHCl

Light source frequency: DC



Analyzing using imagej

Calculations:

We know,

$$HB = \frac{2P}{\pi D [D - (D^2 - d^2)^{\frac{1}{2}}]} \quad (\text{Sundararajan, 2001})$$
$$= 13.89 \text{ kgf/mm}^2$$
$$= 136.21437 \text{ MPa} \quad (1 \text{ kgf/mm}^2 = 9.81 \text{ MPa})$$

Here,

$$P = 500 \text{ kgf}$$
$$D = 10 \text{ mm}$$
$$d = 6.37 \text{ mm}$$

Determination of strength of Al alloy from hardness:

$$YS = 0.383 \cdot HB - 182.3$$
$$= 130.13 \text{ MPa}$$

$$UTS = 0.247 \cdot HB + 113.1$$
$$= 146.74 \text{ MPa}$$

From the graph, we see that,

$$YS = \frac{2.024 \cdot 10^3}{\pi \cdot (5.325 \cdot 10^{-3})^2 \cdot 10^6} = 22.72 \text{ MPa}$$

$$UTS = 75.1435 \text{ MPa}$$

We know,

$$UTS = \sigma_{gb} + \sigma_p + \sigma_{ss} + \tau_D$$

now, $\sigma_{gb} = \sigma_{\alpha} + k_y d^{-1/2}$

The value of 'd', the average grain size, which can be calculated by analyzing the microstructure image. As our

Given that,

$$\sigma_{\alpha} = 60 \text{ MPa}$$

$$K_y = 0.12 \text{ MPa}/\sqrt{m}$$

d =

provided image was not clear, d can not be calculated from that image which was discussed later on.

Again,

$$\sigma_p = 362.7 f_1^{1/2} r_1^{1/2} + \frac{0.1}{\pi} \frac{M G b}{\sqrt{1-v}} \frac{\ln\left(\frac{4r_2}{b}\right)}{r_2 \left(\sqrt{\frac{\pi}{4f_2}} - 1\right)}$$

$$= 323.4238 \text{ nm}^{0.5} - 2.127 \text{ GPa} \cdot \text{nm}$$

Given that,

f1=0.513
r1=1.55nm
f2=7.75
r2=3.75nm
M=3.06
G=26.9 GPa
b=0.499nm
v=0.33

Again,

$$\sigma_{ss} = \sum A C_0^{2/3}$$

where C_0 is the mass percentage of solute atoms. We didn't take OES test for Al alloy.

Again,

$$\lambda = 4r_2 \left(\sqrt{\frac{\pi}{4f_2}} - 1 \right)$$

$$= -10.225 \text{ nm}$$

Here,
r2=3.75nm
f2=7.75

$$\text{Again, } \tau_D = \frac{B G b}{\lambda} = \frac{0.2 * 26.9 * 0.499}{-10.225} \text{ GPa} = -0.2626 \text{ GPa}$$

B=0.2 for FCC material

Result & Discussion:

From our calculations,

From hardness value,

YS(yield strength)=-130.13MPa

UTS(Ultimate tensile test)=146.74MPa

From tensile test graph analysis,

YS=22.72MPa

UTS=75.1435MPa

1)The hardness value should be greater than 20 and less than 200, but it was 13.89 in our case.

That's why we got a negative YS while YS from HB.

2)There's a huge difference between the values that came from the hardness test and from the tensile test graph. We think these differences occurred as we did the tensile test on our first day, We took a small part and repeatedly did polishing and etching until the microstructure came out. So that particular part may get damaged and eventually lose its actual hardness value. Thus a huge deviation from the actual value occurred.

3)While conducting the tensile test using the UTS machine, we observed elongation, but necking was not specifically seen. The specimen fractured abruptly from the mid-region. We then took out the part and measured its elongation. After that, we cut a simple piece of 15mm from the end of our specimen. We then took it to the microstructure analysis lab to do its microstudy. First, we did 150 emery paper grinding and continued to do the grinding up to the finest paper.

4)We conducted the test 5 times on 5 different days on both sides of our specimen. Because of polishing so many times, the surface was corroded, and it showed as a black spot in our microstructure.

5)There was some porosity in our sample. By asking the metallography lab instructor, he told us that Al is softer than mild steel or other alloys, so if you do emery paper grinding multiple times in a single spot, the abraded particles can create porosity.

6)On our first day, we mistakenly used nitrile instead of acetone to dry the surface. That's why one side of our sample got oxidized.

7)After doing it properly and following all the steps line by line, 5 times, we were unable to bring out the microstructure from which we can calculate the average grain size, thus σ_{ss} .

8)From this experiment we have come to know, how to deal with Al samples properly. While wet polishing we have to be careful not to take out the sample from the polishing paper because oxide can be formed so fast in the Al sample's surface.

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